

Finally 'surface'-ing

Microsoft's gigantic coffee table sized touchscreen computer aka "Surface" is set to transition into our living rooms three years from now

First person shooter

The Looxcie is a wearable video camera that can upload the last 30 seconds of your life directly to Youtube at the push of a button

NVIDIA GeForce GTS450

Forcing their way into the crowded space



MSI NGTS450 Cyclone



Galaxy GTS450 GC version



ASUS ENGTS450 Top

NVIDIA's GeForce GTS450 fits into the lower mid-range bracket. There have been some drastic changes in the ingredients of the recipe – a big drop to 192 CUDA cores as compared to the 336 of the GTX460 GPU, but a somewhat faster core. The cards come with 1 GB of GDDR5 memory with a 128-bit memory interface.

Each one of the three cards came factory overclocked with custom coolers.

The Galaxy GTS450 GC version came with a removable fan design which makes cleaning the card simpler. The same design can be found on some of the other Galaxy cards. The MSI N450GTS Cyclone card came with a fancier heatsink design but with nothing covering most of the card. Even the ASUS card had a similar design but with a hood covering the heatsink.

All the cards ran cool in all the tests. On idle, the ASUS and MSI card dropped to under 40°C whereas under maximum load, they stayed under 50°C in a moderately air-conditioned room. The MSI was the coolest at just 46°C under load which was very impressive for graphics cards.

Just the overclocked GPUs aren't enough to make up for the drop in CUDA cores. We noticed a drop of some 15 to 20 per cent in frame rates in games such as Crysis Warhead. The fastest of the overclocked cards was the ASUS ENGTS450 Top. The MSI NGTS450 Cyclone was just a couple of frames per second slower than the ASUS ENGTS450 in almost all of the benchmarks.

The ASUS is the most expensive at ₹9,500, the MSI card ₹8,900 and the Galaxy card at ₹8,500.

As a gaming card, all of these cards are capable of providing very playable frame rates at resolutions up to 1920 x 1200. It's only when you run games such as Crysis Warhead that you notice the performance dropping well under the 30 fps mark, even with the resolution set at 1280 x 1024. You'll definitely have to reduce quality settings drastically to get playable frame rates in Crysis Warhead.

If you are looking for a cheap gaming card in the range of ₹8,000 to ₹9,000, this should be the card to go for. The cards are priced and perform like the AMD Radeon 5750 cards. The Galaxy GTS450 GC version maybe the cheapest but the frame rates in the demanding games is marginally lower. If you want every bit of performance, and don't mind spending that little more, then the ASUS ENGTS450 TOP is undoubtedly the one for you.

Scores and contact details

Brand	ASUS	Galaxy	MSI
Model	ENGTS450 Top	GTS450 GC version	NGTS450 Cyclone
Price	₹9,500	₹8,500 / ₹7,500 (MOP)	Rs. 8,900
Benchmarks			
3DMark Vantage (GPU)	9944	8940	9335
Crysis Warhead (1680 x 1050)	22	19	20
Company Of Heroes: ToV (1680 x 1050)	77.4	73.33	78.3
STALKER Clear Sky (Day, 1680 x 1050)	28.9	25.9	26.8
Ratings			
Features	7.5	7.5	7.5
Performance	7.5	7	7
Build Quality	7.5	7	7
Value for Money	7	7.5	7.5
Overall	7.5	7.5	7.5
Contact			
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